Frozen Yet Adaptive: Single-Episode Repair of Vision–Language–Action Policies from Proprioceptive Residuals

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Abstract

A vision–language–action (VLA) policy is a large frozen network whose training assumed a healthy robot. When the execution hardware drifts—a miscalibrated tool frame, a stiff or weakened joint—the policy is silently wrong, and fine-tuning it is out of reach for most deployments. We show that the fault can be identified and cancelled within a single episode, from the robot's own motion, by adapting six numbers at the policy's action interface: a plant model and a sensitivity matrix identified once on healthy data turn the proprioceptive residual into an estimate of the fault, and a projected, normalised gradient law drives it to zero. No gradients through the policy, no reward, no reference trajectory, and no learning across episodes. On four LIBERO suites with a frozen $\pi_{0.5}$, paired episodes go from 23% to 65% success (exact McNemar $p = 6.1 \times 10^{-13}$, $n = 120$), with every suite individually significant; across 690 paired episodes spanning additive, structured, time-varying and multiplicative faults the correction fixes 275 episodes and breaks 7. The same law and the same calibration repair two further backbones from two other developers (OpenVLA-OFT, NVIDIA GR00T N1.7) and, in joint space, a second manipulator (bimanual ALOHA). A tuned integral baseline without the plant model reaches 35% where the method reaches 95%. We state the two conditions the evidence supports—an identifiability condition and a repairability criterion comparing the task's spatial margin against the correction's variation—and the failure modes that bound the method: faults invisible to proprioception, tasks whose margin is below the estimator's stationarity, and a policy that cannot do the task unfaulted.

1 Introduction

Vision–language–action models map images and a language instruction to robot actions and are trained on demonstrations collected from a particular robot in a particular state of repair. Deployed, they are frozen. The robot is not: tool frames drift, calibrations go stale, joints lose effectiveness. Each of these changes the map from the policy's action to the robot's motion, and the policy—which never observes its own commanded motion against the achieved one—has no mechanism to notice.

The standard remedies are to fine-tune on faulted data, which requires labelled faults and the compute to update billions of parameters, or to learn a residual correction offline against reference rollouts. Both need the fault to be represented in a dataset before it can be corrected—the closest contemporary method, J-PARC (Jo et al., 2026), trains its calibrator per environment from faulted rollouts against fault-free references and teacher targets. Neither is available to a robot that degrades while running.

This paper takes the classical control view. A fault at the action interface is a disturbance on a plant whose healthy dynamics can be identified once; the disturbance is observable in the difference between the motion the healthy plant would have produced and the motion actually measured; and an adaptive law with a handful of parameters can cancel it online. The question is whether this survives contact with a 3.35B-parameter policy, contact-rich tasks, and a residual that carries model error as well as the fault. Our contributions are the answer:

1. A single-episode repair loop for frozen VLAs (§4) needing only a plant model and a sensitivity matrix identified on healthy data, both properties of the robot rather than of the policy or the task. The estimate resets every episode and reconverges in about 15 control steps.
2. Evidence at scale, paired (§6): four suites, three backbones, two manipulators, four fault families, three severities and four time profiles, evaluated episode-for-episode against the frozen policy with exact McNemar tests, a healthy-frozen control on every cell, and 690 paired episodes in aggregate.
3. Two conditions, stated as propositions (§5): a constant input fault and a constant output bias are not separable within one episode, so the healthy bias must be measured outside it; and repair succeeds when the correction's within-episode variation fits inside the task's spatial margin, both quantities being measurable before any repair attempt. The second predicts every outcome on a second manipulator, including the one continuous adaptation fails.
4. The boundary (§7): what the method cannot see (sensor and camera faults), cannot fix (a policy that fails the task unfaulted), and gets wrong (one healthy episode in twenty on one backbone, caused by a sustained phantom).

We also report, rather than hide, that three of our own mechanistic explanations were refuted by our own measurements during this work; the surviving claims are the ones that outlasted them.

2 Related work

VLA policies and their fragility. π_0 and $\pi_{0.5}$ generate action chunks by flow matching from a vision–language backbone; OpenVLA and its OFT variant use an autoregressive backbone with a regression head. Both are evaluated on LIBERO in a fixed simulated robot. Robustness to execution faults is not part of the evaluation protocol of either.

Residual calibration of frozen VLAs. The closest work is J-PARC (Jo et al., 2026), which studies joint-level physical faults—joint lock, limited range of motion, increased friction—on frozen $\pi_{0.5}$ and OpenVLA-OFT across the same four LIBERO suites, and reports a WidowX demonstration. It trains a lightweight residual calibrator offline, per environment, from faulted rollouts paired with fault-free reference rollouts and teacher-oracle residual targets, and reports 48.4% → 55.2% on $\pi_{0.5}$ and 42.3% → 46.8% on OpenVLA-OFT under joint lock (unpaired, 500 episodes per suite). The differences define our contribution. We use no faulted data, no reference trajectories and no teacher: the plant is identified once on healthy motion and the fault is identified online within the episode in which it occurs, on paired episodes (23% → 65%). Their faults, however, enter below the controller, where the fault-to-motion map is state dependent; ours enter at the action interface above it, and the joint-level class appears in this paper only in the hardware protocol (§6.7). The two are complementary on exactly that axis. Closed-loop affine activation editing (Das et al., 2026) steers a frozen policy by editing internal activations against an observed objective; our correction lives outside the network and needs no access to activations.

Adaptive and fault-tolerant control. Our law is a disturbance observer with feedforward cancellation, in the MRAC gradient form $\dot{\theta} = -\Gamma\phi e$, with the standard robustness modifications (deadzone, normalisation, projection). What is new is not the law but its target: a frozen generative policy whose command distribution determines what is identifiable, and a task whose spatial margin determines what is repairable.

3 Setting

A frozen policy π_ω emits, at 20 Hz (LIBERO) or 50 Hz (ALOHA), an action chunk from which the controller consumes $a_t \in \mathbb{R}^d$ per step. For LIBERO a_t is a six-dimensional

Cartesian increment plus gripper, executed by an operational-space controller; for ALOHA it is fourteen absolute joint targets executed by position servos. The measured state y_t is the end-effector increment (LIBERO) or the joint position (ALOHA).

A fault corrupts the executed action:

$$a_t^{\text{exec}} = a_t + f \quad \text{(additive)}, \qquad a_t^{\text{exec}} = \operatorname{diag}(g)\, a_t \quad \text{(multiplicative)},$$

constant or varying in time. The fault is injected between the policy and the controller. For LIBERO this models a miscalibrated tool frame or a stale hand–eye calibration; it is not a joint-level actuator fault, which would enter below the controller where the fault-to-motion map is state dependent (§8).

The policy never sees f. It sees images and the proprioceptive state, and $\pi_{0.5}$-LIBERO in particular ignores the state input entirely (zeroing it changes the action by exactly 0.00000), so a perception-side fault cannot be constructed for it and an action-side fault cannot be corrected by it.

4 Method

4.1 Plant, residual, and sensitivity

On healthy rollouts we identify a per-channel FIR plant

$$\hat{y}_t = \sum_{\ell=0}^{K} h_\ell\, u_{t-\ell} + c, \qquad K = 6, \tag{1}$$

where u_t is the command we believe we sent (the policy's action plus any correction we applied). For LIBERO the plant predicts the increment; for ALOHA it predicts the position, because there the command is an absolute target that a servo tracks and a target offset lives in the steady-state position error, not in the increment (§6.6). The residual

$$r_t = y_t - \hat{y}_t \tag{2}$$

is what the healthy plant cannot explain. Because u_t includes the correction and the world executes $u_t + f$, $r_t \approx Mf$ independently of the correction, so the residual estimates the total fault. $M = \partial(\text{motion})/\partial(\text{fault})$ is a $d \times d$ sensitivity matrix measured once by open-loop replay: a recorded healthy command sequence is replayed with and without a per-axis probe, and the mean motion difference per unit probe gives each column. On the LIBERO plant M is diagonal-dominant with $\operatorname{cond}(M) = 3.0$ and is magnitude independent to within 14% between probes of 0.02 and 0.05; on ALOHA $M \approx I$ with $\operatorname{cond}(M) = 7.3$.

4.2 The adaptive law

Per control step,

$$e_t = M^{-1} r_t - b, \qquad \tilde{e}_t = \frac{e_t\, \mathbb{1}[\|r_t\| > \delta]}{1 + \|r_t\|^2/\rho^2}, \qquad \hat{f}_{t+1} = \Pi_{[-\kappa,\kappa]}\big(\hat{f}_t + \gamma(\tilde{e}_t - \hat{f}_t)\big), \tag{3}$$

and the next action is sent as $a_t - \hat{f}_t \odot m$, where m selects the channels to correct. The terms are the standard robustness modifications and each earned its place by fixing an observed failure: the deadzone δ stops plant noise integrating into drift, the normalisation damps outlier steps (contacts, joint limits), the projection κ keeps the estimate in a physical box, and b is the healthy-plant phantom—the value $\hat{f}$ settles at with no fault present—measured offline and subtracted, exactly as a sensor is zeroed before it is trusted (Proposition 1 says why it cannot be fitted within the episode).

Remark 1 (Constants are ratios of measured scales). Every threshold in equation 3 is compared against a quantity with a scale: δ against the residual norm of the healthy plant, ρ against the residual norm under the fault, κ against the expected fault, the excitation gate of §4.4 against the regressor's magnitude. Five silent failures in this work came from carrying a constant tuned on one plant to another—a deadzone larger than the residual it gated, a normaliser that attenuated every ALOHA update to 6%, a projection bound that coincided with the true fault and was read as a measurement. The rule that follows is mechanical: print the scale, then set the constant as a multiple of it, on every new plant.

The fixed point of equation 3 is attenuated by $1/(1+\|Mf\|^2/\rho^2)$; at the measured $\|r\|$ this is 5%, matching the observed 2% shortfall of the estimate, and an innovation-normalised variant that removes it was measured to be indistinguishable (0.12σ) at twice the variance. We keep equation 3.

4.3 Which channels to correct

The separation test—the estimate under a fault minus the estimate on matched healthy episodes—is the only evidence of identification we accept; a raw $\hat{f}$ carries the plant phantom and cannot distinguish identification from bias (this standard, applied to our own multiplicative law, retracted a claim we had called decisive). For the uniform six-axis fault used across the LIBERO suites, rotation identifies (+0.049 against a true 0.050) and translation does not (sign-wrong), so the headline correction is restricted to rotation. That restriction is a property of that fault, not of the channel: on a translation-only fault the same law identifies translation at 79%–106% and repairs it (§6.3). What makes the uniform fault different is measurable: single-axis responses superpose at ± 0.02 and stop superposing at ± 0.05, asymmetrically (x: 0.07 in one direction, 0.78 in the other), the signature of a contact or joint limit reached only when all six axes are perturbed together.

4.4 Multiplicative faults

For a gain fault the regressor follows from the plant rather than being assumed. The world executes $g\,u$ and the FIR responds to what it is given, so per channel

$$y_i - \hat{y}_i = \beta_i\,\phi_i, \qquad \phi_i = \hat{y}_i - c_i, \qquad \beta_i = g_i - 1, \tag{4}$$

the regressor being the FIR-weighted command history, and the regression lives in motion units with no M^{-1}. Estimating β_i by recursive least squares on channels whose $|\phi_i|$ clears an excitation gate, the correction is the exact certainty-equivalent inverse $a_i/(1+\hat{\beta}_i)$ with $\hat{g}_i$ projected away from zero. Two facts matter here. The compensator inverts the estimate, so estimation error is amplified by $1/\hat{g}^2$—fourfold at $g = 0.5$—and accuracy sufficient for an additive fault is not sufficient for a multiplicative one. And a wrong regressor is not repaired by adding parameters: regressing on the instantaneous command instead of ϕ_i produced a 33% phantom loss of effectiveness on a healthy robot, an intercept made it worse, and the derived regressor shrank the phantom ninefold and took the x-channel separation from 1.4σ to 18σ.

4.5 Identify, then hold

On a task whose spatial margin is smaller than the correction's within-episode variation, continuous adaptation fails even when its mean is inside the margin (§6.6). The scheme the evidence supports is to adapt during one sacrificial episode and hold the estimate thereafter: a persistent hardware fault gets a persistent estimate, at the cost of one episode, which is what a calibration procedure costs anyway.

5 Two conditions

Proposition 1 (A constant input fault and a constant output bias are not separable). Let $y_t = M(a_t + f) + b + \nu_t$ with M square and full rank, and f, b constant over the episode. For any b', the fault $f' = f + M^{-1}(b - b')$ reproduces every observation exactly. Only $Mf + b$ is identified.

Proof. $M(a_t + f') + b' = Ma_t + Mf + (b - b') + b' = M(a_t + f) + b.$ $\square$

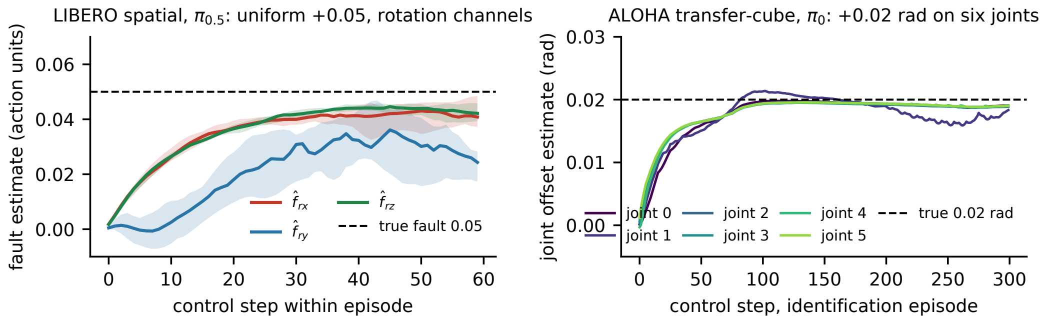

Figure 1: The fault estimate within one episode, from zero. Left: LIBERO spatial with a frozen $\pi_{0.5}$ under a uniform +0.05 fault on all six action dimensions; the three rotation channels (median and interquartile range over 20 episodes) with the correction on rotation only: two reach 0.04 by step 60 and the third 0.024, all still rising; at episode end the separation test puts them at 0.047 to 0.049 (§4). Right: ALOHA transfer-cube with a frozen π_0 under a +0.02 rad offset on the left arm's six joints; the joint-space estimate in the identification episode settles at 0.0183 to 0.0191 rad, 92 to 96% of the true offset, and is then held for the remaining episodes (§6.6).

Consequently the healthy phantom b must come from outside the episode (we measure it on fault-free rollouts), and a fault that switches on mid-episode is better posed than one present from the start: b is identified before onset and f after. Mid-episode onset is a well-posedness condition, not a robustness flourish.

Proposition 2 (Repairability). Let μ be the task's spatial margin—the largest steady residual offset at the end-effector under which the healthy policy still succeeds at its healthy rate—and let σ be the within-episode variation of the applied correction. Repair recovers the healthy rate when the residual stays within μ for the whole episode, for which $\sigma < \mu$ is necessary. Both μ (by a static-correction sweep) and σ (from the estimator's trajectory on one faulted episode) are measurable without knowing the fault and without a repair trial.

This is an empirical proposition, not a theorem: it states the quantity the data turned out to depend on. It predicts LIBERO (a reach with centimetres of margin; σ of 0.019 cm per step; repair), ALOHA with continuous adaptation ($\mu < 0.5$ cm; $\sigma = 0.43$ cm; no repair), and ALOHA with the held estimate ($\sigma = 0$; repair) (§6.6). A stronger, contraction-based recovery bound would need the composite policy–controller–robot map to contract on a task tube, which is not established for a contact-rich task, and we do not claim it.

6 Experiments

6.1 Protocol

Paired design. Both arms of every cell run the same (task, initial state, seed) episodes and differ only in whether the correction is applied; Figure 1 shows what the corrected arm does inside one episode. We report exact McNemar tests on the discordant pairs; unpaired tests on these data are conservative and, worse, hide the regression count. Controls. Every cell has a healthy-frozen control on the same episodes: a null against a ceiling (frozen already near 100%) or a floor (frozen near 0 unfaulted) is not a null, and four cells in this work were avoidable with that control run first. Noise. $\pi_{0.5}$ samples its action flow; the identical condition scored 13/20, 13/20, 15/20 on three runs, so ±10 points at $n = 20$ is free variation and we quote intervals. Calibration. Plant and M were identified once on libero_spatial healthy rollouts with $\pi_{0.5}$ driving and are used unchanged for every LIBERO suite and for OpenVLA-OFT; ALOHA has its own, from eight healthy rollouts.

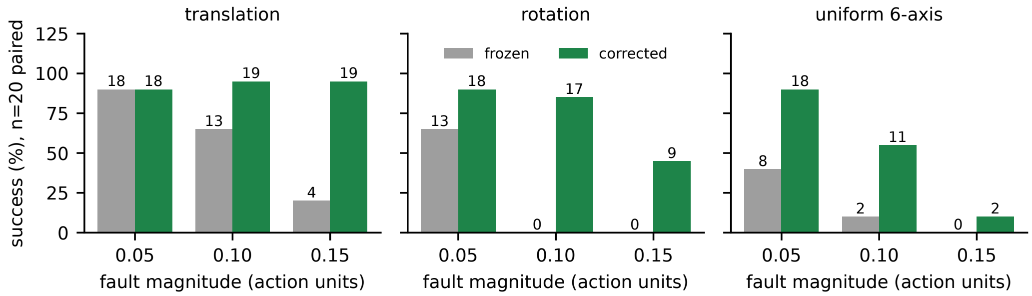

Figure 2: The recoverability map: frozen versus corrected success on LIBERO spatial, $n = 20$ paired episodes per cell, correction on the channels each family's separation test supports (rotation for the rotation and uniform families, translation for translation). No cell has a regression. Single-family faults degrade gracefully with magnitude; the uniform fault collapses at 0.15 because its uncorrected translation component drives the arm into contacts and limits (§7). Exact McNemar p-values are in the table below.

6.2 Four suites

Uniform +0.05 offset on all six action dimensions, rotation-only correction (§4.3), $\pi_{0.5}$-LIBERO.

suite	n	frozen	corrected	fixed	broken	exact McNemar
libero_spatial	20	8/20 (40%)	18/20 (90%)	10	0	0.0020
libero_goal	40	15/40 (38%)	29/40 (72%)	15	1	0.00052
libero_object	20	5/20 (25%)	16/20 (80%)	11	0	0.00098
libero_10	40	0/40 (0%)	15/40 (38%)	15	0	6.1×10^{-5}
pooled	120	28/120 (23%)	78/120 (65%)	51	1	$\mathbf{6.1 \times 10^{-13}}$

Every suite is individually significant. The long-horizon suite goes from a frozen policy that never succeeds in forty episodes to fifteen recoveries. The single regression is a goal episode the frozen policy itself solves three times in four, with an ordinary-sized correction. libero_90, on which the checkpoint was never fine-tuned, scores 6/20 unfaulted and is reported as a competence floor of the policy, not a repair result.

6.3 Fault families and severity

Figure 2 and the table give the fault-family by magnitude map.

fault (libero_spatial, $n = 20$)	0.05	0.10	0.15
translation	18/20 → 18/20 (ceiling)	13/20 → 19/20	4/20 → 19/20
rotation	13/20 → 18/20	0/20 → 17/20	0/20 → 9/20
uniform 6-axis	8/20 → 18/20	2/20 → 11/20	0/20 → 2/20

Zero regressions in every cell. Rotation is where this policy breaks—a rotation fault at 0.10 takes it to exactly zero where translation leaves 13/20—and the correction recovers 17/20 from that floor. Single-family faults degrade gracefully with severity; the uniform fault collapses, and the comparison of the last column isolates why: the rotation fault and the uniform fault are corrected identically, and adding an uncorrected translation component drops recovery from 45% to 10%. The boundary is the plant leaving the regime where its motion is informative, not the estimator. A structured mixed-sign fault $[0, 0, 0, +.06, -.06, +.03]$ the law was never tuned on goes $2/12 \rightarrow 10/12$ with separation matching truth on every axis.

Time-varying faults (amplitude 0.10): a ramp to full over 60 steps goes $12/20 \rightarrow 18/20$ ($p = 0.031$); a non-zero-mean oscillation on rotation $13/20 \rightarrow 20/20$ ($p = 0.016$); an intermittent fault 12/20 $\rightarrow$ 19/20 ($p = 0.039$). The estimator tracks the waveform poorly—45–79% RMS error, not attributable to lag—and it does not matter: the correction must be roughly right in direction and scale, not a good tracker, which is why six numbers updated on CPU suffice. (A zero-mean oscillation averages away and does no damage; that cell proved nothing and is not counted.)

Loss of effectiveness (§4.4): at $g = 0.30$, 1/20 $\rightarrow$ 17/20 ($p = 3.1 \times 10^{-5}$) with $\hat{\beta} = (-0.723, -0.699, -0.717)$ against -0.700; at $g = 0.20$, 0/20 $\rightarrow$ 17/20 ($p = 1.5 \times 10^{-5}$). At $g = 0.5$ the frozen policy still scores 15/20 and the cell is a ceiling. Rotation gain faults are unidentifiable for this policy: no recorded step clears the excitation gate on r_x or r_z at any threshold, because the policy does not rotate the wrist. That is a property of the policy's behaviour, not of the estimator.

6.4 Baselines

Frozen and a static oracle correction equal to the true fault bound the problem. The informative baseline is the one that asks whether the plant model and M earn their place: integral action on the raw motion error, $\hat{f} \leftarrow \hat{f} + k_i(y_t - a_t)$, swept over five gains across two orders of magnitude and reported at its best. On the 0.15 translation fault:

k_i	frozen	baseline
0.001	4/20	5/20
0.005	3/20	7/20 (35%)
0.02	5/20	0/20
0.15	4/20	1/20
ours	4/20	19/20 (95%)

The baseline implicitly assumes achieved motion equals commanded action; the measured plant DC gain is 0.23, so with no fault at all its error is $-0.77\,a$—at a command of 0.5 a phantom of 0.384 against a real fault of 0.150, 2.5× the target and opposite in sign. It must creep to stay stable and saturates when it moves. Predicting $y \approx 0.23a$ instead of $y \approx a$ is the whole difference.

6.5 Two further backbones, unchanged calibration

OpenVLA-OFT (7B, autoregressive backbone with an L1 action head) and NVIDIA GR00T N1.7 (3B, Cosmos VLM with a flow-matching DiT head, the official LIBERO finetune) are each served behind the same interface; every script, the plant and M are the $\pi_{0.5}$ ones.

cell (libero_spatial)	frozen	corrected	fixed	broken	McNemar
healthy control	20/20	19/20	0	1	1.0
rotation 0.10	0/20	11/20	11	0	9.8×10^{-4}
translation 0.15	0/20	14/20	14	0	1.2×10^{-4}
gain 0.20	0/20	17/20	17	0	1.5×10^{-5}

Nothing in the method was $\pi_{0.5}$-specific. OFT is more fragile (frozen 0/20 on all three faults), recovers less on rotation (55% vs 85%; its r_y fault is only 23% identified where $\pi_{0.5}$ reached 87–100%), and the healthy control lost one episode to the law itself: a sustained phantom at four to ten times the usual size, which a confidence gate evaluated offline does not remove without delaying genuine repair by one to two seconds (§7). The three headline cells also replicate on libero_object with the same calibration (0/20 $\rightarrow$ 10/20, 0/20 $\rightarrow$ 15/20, 0/20 $\rightarrow$ 19/20).

GR00T N1.7 cell (libero_spatial)	frozen	corrected	fixed	broken	McNemar
healthy control	18/20	18/20	1	1	1.0
rotation 0.10	0/20	14/20	14	0	1.2×10^{-4}

GR00T is a third developer and a third architecture. Its healthy control is a null with a rotation phantom below 0.005; a rotation fault that zeros it is recovered to 70% with no retuning, the estimate settling at 81%, 42% and 82% of the true fault on the three axes. The under-identified r_y is the same channel that lags on the other two backbones: it is a property of the plant and M, not of the policy. Across the three backbones on this cell, 42 of 60 paired episodes are repaired and none is broken.

6.6 A second manipulator, in joint space

Bimanual ALOHA in gym_aloha driven by π_0: fourteen absolute joint targets at 50 Hz, transfer-cube, 300-step episodes, healthy success 25%. The plant is identified on position from eight healthy rollouts ($R^2 = 0.989$–1.000 on all joints), $M \approx I$. A +0.05 rad offset on the six left-arm joints displaces the gripper by 5.3 cm; the frozen policy scores 0/20 at every severity tested.

Identification transfers. Separation against the healthy control: 85–91% of the fault at 0.05 rad, 94–106% at 0.02 rad, leakage ≤ 0.007 rad onto uncorrected joints. Repair does not, at first. With continuous adaptation the task returns 0/20 at every severity, warm-started or cold, with or without an early-history defect we found and fixed. The reason is quantitative:

correction (0.02 rad fault, $n = 20$)	within-episode variation	mean residual	success
none (frozen)	—	2.12 cm	0/20
adaptive, updating throughout	0.43 cm	0.17 cm	0/20
0.019 rad (95%) applied frozen	0	0.11 cm	5/20
adapt episode 0, then hold	0	0.12 cm	5/20 ($p = 0.0$
adapt episode 0, then hold, $n = 40$	0	—	15/40 ($p = 6.1 \times 10^{-}$
0.020 rad exact, from step 0	0	0	8/20
healthy, no fault	—	—	5/20
healthy, no fault, $n = 40$ (law running: 14/40, $p = 0.58$)	—	—	17/40

A static-correction sweep puts the task's margin under 0.5 cm: a 90% correction held for the whole episode restores nothing, 100% restores the healthy rate. The updating law reaches a mean residual well inside that margin and still fails, because its correction moves by 0.43 cm within an episode—the whole margin—on an arm that is precise from step one. The same number, applied stationary, repairs. Adapting for one episode and holding the estimate recovers 5/19 on the held episodes at $n = 20$, the healthy-frozen rate, and 15/39 at $n = 40$ (exact McNemar $p = 6.1 \times 10^{-5}$, zero regressions; held estimate 0.019 to 0.020 rad, 95 to 101% of the offset), from a frozen policy at zero. The healthy arm on the same seeds scores 17/40, and the repaired arm is indistinguishable from it ($p = 0.75$); the law on a healthy arm holds a phantom of at most 0.0008 rad and changes nothing (14/40 against 17/40, $p = 0.58$). This is the content of Proposition 2, and it was invisible on LIBERO because a 20 Hz Cartesian reach tolerates centimetres of jitter.

6.7 Hardware

The same law was applied to logs from an X2 humanoid running a learned box-pickup policy, in which a real actuator-authority fault (gain_scale, scaling stiffness, damping and feedforward together) spans 0.1–1.3 across 134 rollouts. Three findings fix the protocol for the repair experiment that follows. The fault is absorbed by the position loop and is invisible in joint position, but recoverable in effort ($r = +0.84$). The plant is run-varying on the load-bearing joints, so the calibration window must be within-run and must contain motion (≥ 450 steps), not the standing phase. The detectability floor is 0.5–1.3° of joint offset. The injection knob for a joint-target offset is in the deploy script; the paired repair experiment on the robot is pending robot time and is the one result this paper does not yet contain.

7 What does not work, and what was wrong

Invisible faults. A proprioceptive bias and a wrist-camera misalignment both damage the policy (6/12 for the camera shift) and neither reaches the residual (separation 0.004 against 0.049 for an action fault). The method sees the action interface and nothing upstream of it.

A policy that cannot do the task. libero_90 fails 14 of 20 sampled tasks unfaulted. Correcting the fault restores nothing because nothing was lost to it.

Harm on a healthy robot. One OFT episode in twenty was lost to a sustained phantom integrated and acted on. A dwell gate cuts spurious openings from 13/20 to 4/20 healthy episodes but the harmful one opens at every setting tried, and every setting delays real repair. Separating it would need information the estimator does not have.

Continuous adaptation on a tight margin (§6.6).

Refuted explanations, kept. We proposed that identifiability is predictable from healthy action statistics via the quantile scale (refuted: translation identifies a single-axis fault at 79%); that the plant is nonlinear on translation (refuted: M is magnitude independent to 14%; superposition fails instead); and that translation correction contributes nothing (refuted: it was tested only where the policy shrugs off a translation fault; at 0.15 it takes 20% $\rightarrow$ 95%). A fourth, that additive and multiplicative faults are identifiable on complementary channels, survives only as a graded ordering. The record keeps each with the measurement that killed it.

8 Limitations

The LIBERO fault is at the Cartesian action interface, above the controller; a joint-level actuator fault—the class J-PARC studies—enters below it with a state-dependent map, and the hardware section is the only place that class appears here, without a repair result yet. Whether the residual of a position loop carries a joint lock or friction change as cleanly as it carries an offset is open; our X2 logs say a stiffness loss is invisible in position and recoverable in effort, which suggests the sensor, not the law, is what changes. The estimate resets every episode by design for the LIBERO claims; the ALOHA result shows that on a tight-margin task the deployable scheme is instead to hold it, which is a different claim about online adaptation and is stated as such. The policy's stochasticity gives ± 10 points at $n = 20$; we mitigated it with pairing and $n = 40$ where it mattered, not by more seeds. We give propositions, not a recovery theorem; a contraction argument on the composite map is the natural next step and requires assumptions we could not verify on contact-rich tasks. And every constant in the law is a ratio of a measured scale that must be re-measured on a new plant—five of our own failures came from not doing so.

9 Conclusion

A frozen VLA can be repaired within one episode by six numbers at its action interface, given a plant model and a sensitivity matrix identified once on healthy data. The evidence is paired, controlled, and broad; the calibration transfers across suites and backbones; the two conditions that govern success are stated and measurable in advance; and the failures are named. The result the paper still owes is the one on a real robot.

AI use statement

Generative AI (Claude, Anthropic) was used throughout this work as a coding and writing assistant under the authors' direction: it wrote and edited the experiment scripts, the analysis code, the running project record from which this paper is built, and this manuscript's text; the authors set the research questions, chose the experiments, ran or supervised every run on their own hardware, and reviewed the outputs. All numbers in the paper are recomputed from stored per-episode outcomes by a stdlib-only script, so no reported figure rests on model-generated text. Three mechanistic explanations proposed during the work, some AI-generated, were refuted by our own measurements and are reported as such (§7). Related work was verified against the cited preprints. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

Reproducibility statement

Every number above is recomputed from stored per-episode outcomes by openpi/mcnemar.py with nothing installed; the anonymised repository holds those outcomes, the plant and sensitivity calibrations, and every script named in the paper. Experiments need openpi at commit 15a9616, the pi05_libero checkpoint, and one GPU. The two-virtualenv split, the OpenVLA-OFT and ALOHA environments, and the three pins that were not obvious are in SETUP.md. The law's constants are stated as ratios of measured scales (§4), which is what made the calibration transfer across suites and backbones without retuning.

References

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